Customer Purchase Propensity

Data Cleaning & Feature Engineering Pipeline

---

1. Project Overview

This project builds a complete data preprocessing and feature engineering pipeline to prepare an e-commerce dataset for a binary classification Machine Learning model that predicts whether a customer will make a purchase.

2. Techniques Used

Data Sources

- customers.csv — 8 base customers with demographics
- transactions.json — 8 transaction records with product and payment info
- products.sql — 6 products loaded via SQLite
- API (dummyjson.com/users) — 1,000 synthetic users replicating the API schema

Handling Missing Data

- Simple Imputer (median strategy) — age, income, total_spend, num_transactions
- Most Frequent Imputation — gender, city, education, satisfaction, fav_category
- Missing Indicator — binary flag 'income_missing' created before imputation
- KNN Imputer (k=5) — total_purchases using nearest neighbors
- MICE Algorithm (IterativeImputer, max_iter=10) — chained imputation on correlated features
- Complete Case Analysis — reference count (rows remaining if all NaN dropped)

Outlier Detection & Handling

- Z-Score Method (|z| > 3) — income: 11 outliers, total_spend: 11
- IQR Method (1.5 x IQR) — income: 39 outliers, total_spend: 19
- Percentile Method (1st–99th) — income: 22, total_spend: 22
- Winsorization — income, total_spend, total_txn_amount capped at 1st/99th percentile

Date/Time & Mixed Variables

- signup_date and last_purchase_date converted to datetime
- Derived: days_since_signup, days_since_last_purchase, signup_year, signup_month
- Mixed variable phone_raw (e.g. 'C0101-PH') — numeric ID extracted via regex

Encoding Categorical Data

- Label Encoding — gender (Male/Female/Other → 0/1/2)
- One-Hot Encoding — city (18 cities) and fav_category (5 categories)
- Ordinal Encoding — education (5 levels) and satisfaction (5 levels)
- Numerical Binning — income grouped into Low/Medium/High/Very High (quantile-based)

Feature Scaling

- StandardScaler — demonstrated on age, income, total_spend
- MinMaxScaler — scaled to [0,1] range
- MaxAbsScaler — preserves sparsity
- RobustScaler — applied to main pipeline (robust to outliers)
- Normalizer — unit-norm per row
- ColumnTransformer — combined StandardScaler + RobustScaler simultaneously

Feature Construction & Transformation

- purchase_per_day = total_purchases / (days_since_signup + 1)
- spend_per_purchase = total_spend / (total_purchases + epsilon)
- FunctionTransformer: Log, Square Root, Reciprocal on income
- PowerTransformer: Yeo-Johnson and Box-Cox on income
- Equal-width binning of income (5 bins)
- frequent_buyer flag: 1 if total_purchases > 75th percentile

---

3. Biggest Issues With the Raw Data

Issue	Description & Resolution
High Missing Rate (99%)	num_transactions, total_txn_amount, fav_category, payment_modes were missing for API users not present in transactions — resolved via median/mode imputation
Income Missingness (7.9%)	income was deliberately missing for ~8% of API users — resolved with SimpleImputer + Missing Indicator flag
Right-Skewed Distributions	income and total_spend showed strong right skew — resolved via Winsorization + Log/Yeo-Johnson transforms
Mixed Type Column	phone_raw stored IDs in format 'C0101-PH' — numeric part extracted via regex FunctionTransformer
High Cardinality in City	18 distinct cities — handled with One-Hot Encoding after imputing missing cities with mode
Ordinal Categorical Columns	education and satisfaction are inherently ordered — OrdinalEncoder applied with explicit category order

Date Stored as String	signup_date and last_purchase_date stored as strings — converted with pd.to_datetime, derived temporal features

4. Which Imputation & Scaling Worked Best

Best Imputation

- KNN Imputer performed best for total_purchases — leverages multivariate relationships between age, income, and purchase frequency.
- MICE (IterativeImputer) is theoretically superior for correlated features; used on income + total_spend cluster.
- Missing Indicator for income provided the model with information that missingness itself is potentially a signal.

Best Scaling

- RobustScaler was selected for the main pipeline due to the presence of income and total_spend outliers that StandardScaler would over-penalise.
- MinMaxScaler was reserved for neural network input scenarios; ColumnTransformer demonstrated combining both scalers on different feature subsets.

5. Final Dataset Statistics

Total Records	1,008 rows
Total Features	48 columns
Purchased = 1	~55% of records
Purchased = 0	~45% of records
Missing Values (final)	0
Outliers Handled	Winsorized — income, total_spend, total_txn_amount
New Features Created	purchase_per_day, spend_per_purchase, frequent_buyer, income_log, income_sqrt, income_yeojohnson, income_boxcox, income_bin_eq_width, income_group_enc, days_since_signup, days_since_last_purchase, signup_year, signup_month, customer_id_extracted, income_missing

Submitted by: [Nandish Patel|